Chemistry - Undergraduate Level Assessment 21 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. The anhydride of Ba(OH)2 is
   (a) BaH 2
   (b) BaOH
   (c) Ba
   (d) BaO
2. The +1 oxidation state is more stable than the +3 oxidation state for which group 13 element?
   (a) B
   (b) Al
   (c) In
   (d) Tl
3. Reduction of D-xylose with NaBH 4 yields a product that is a
   (a) racemic mixture
   (b) single pure enantiomer
   (c) mixture of two diastereomers in equal amounts
   (d) meso compound
4. The T_2 of an NMR line is 15 ms. Ignoring other sources of linebroadening, calculate its linewidth.
   (a) 0.0471 Hz
   (b) 21.2 Hz
   (c) 42.4 Hz
   (d) 66.7 Hz
5. The 13 C spectrum of which isomer of C_6H_{14} has lines with three distinct chemical shifts?
   (a) hexane
   (b) 2-methylpentane
   (c) 3-methylpentane
   (d) 2,3-dimethylbutane

True / False (10 marks)

Answer True or False

6. True or False: The number of allowed energy levels for the 55 Mn nuclide is 3.

7. True or False: The equilibrium polarization of 13 C nuclei in a 20.0 T magnetic field at 300 K is 1.71 x $10^{-}5$.

7. True or False: Silicon carbide is an n-type semiconductor because it contains more holes than electrons.

8. True or False: The infrared absorption frequency of deuterium chloride (DCl) is shifted from hydrogen chloride (HCl) due to differences in reduced mass.

9. True or False: Silicon most commonly occurs naturally in the form of oxides, such as silica and silicates.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the advantages of using spin trapping techniques for detecting free radical intermediates, particularly in terms of sensitivity and power requirements for detection.

12. Discuss the relationship between the rate of a zero-order chemical reaction and the rate constant, k. How does this relationship influence the reaction's behavior over time?

End of Paper